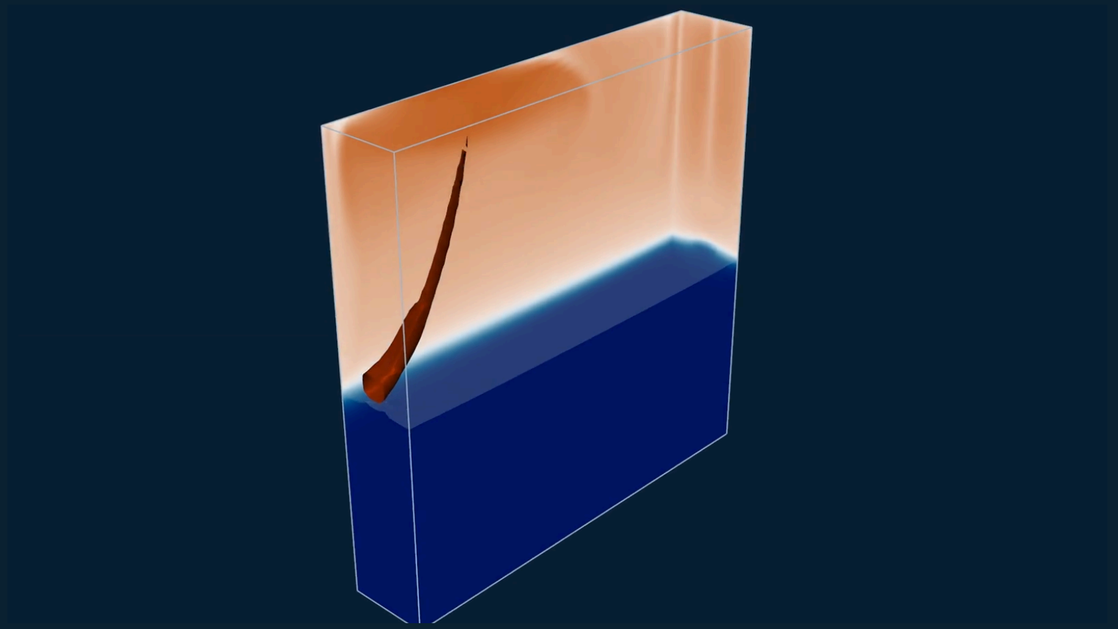
RESEARCH TALK FOR AN INDUSTRIAL PARTNER

The flow settles in seconds. The temperature takes *hours*.

Record once, replay many times: data-based simulation of heat in a buoyant flow



cold plumes under a cooled lid, reference simulation, 3D render of stored solver states

Most of the cost goes into the *flow*, not into the temperature we want.

569 min

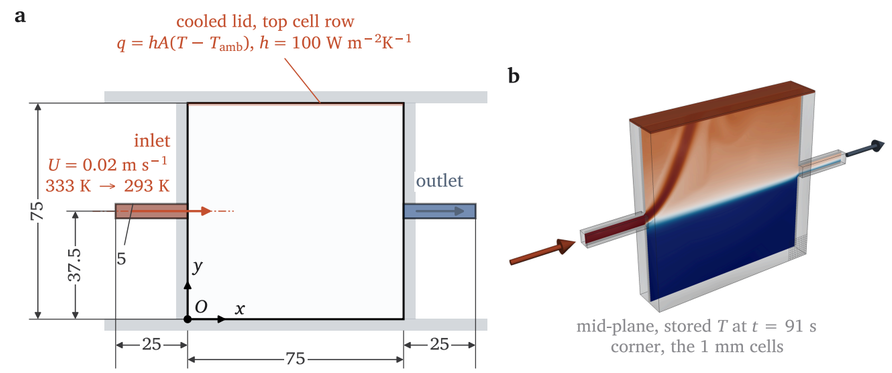
solver time of the reference simulation for about ten minutes of flow in a 75 mm tank

In each time step the solver couples velocity and pressure again, even after the flow has settled into its pattern. The temperature it carries changes far more slowly.

THE ENGINEERING QUESTION

Can we keep the temperature physics and stop paying for the flow every time?

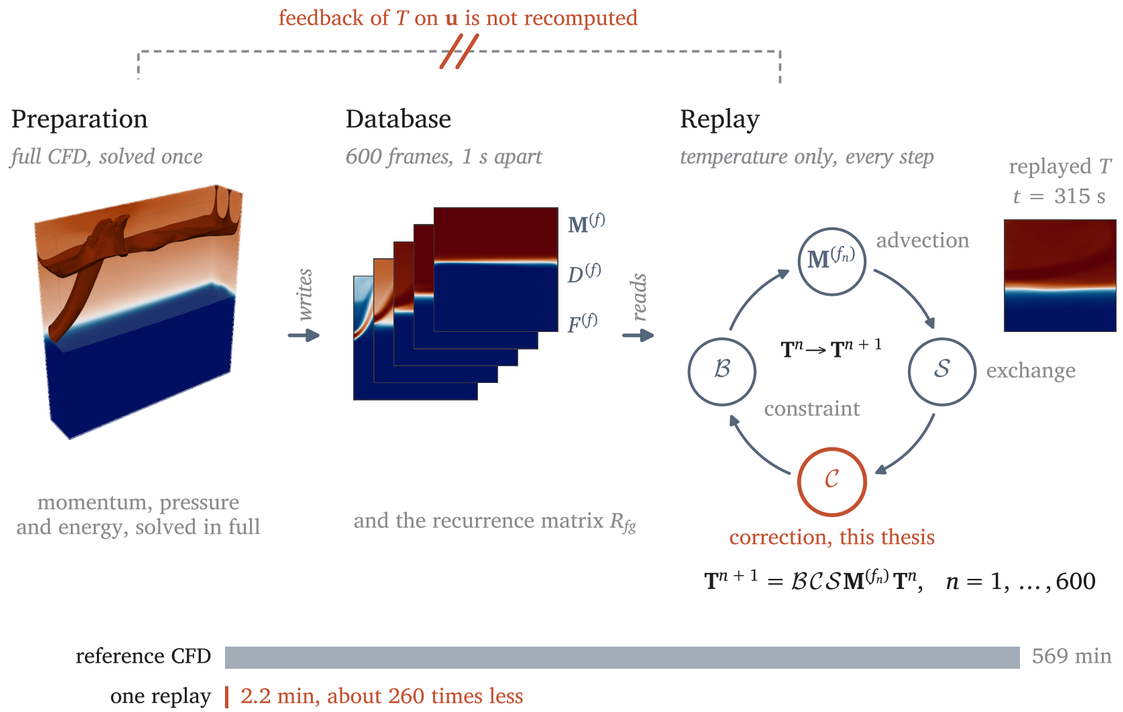
The answer depends on whether the temperature changes the flow. That is what my thesis tests.



the test tank: 75 mm, inlet and outlet channels at mid-height, lid adiabatic or cooled

THESIS FIG 3.2

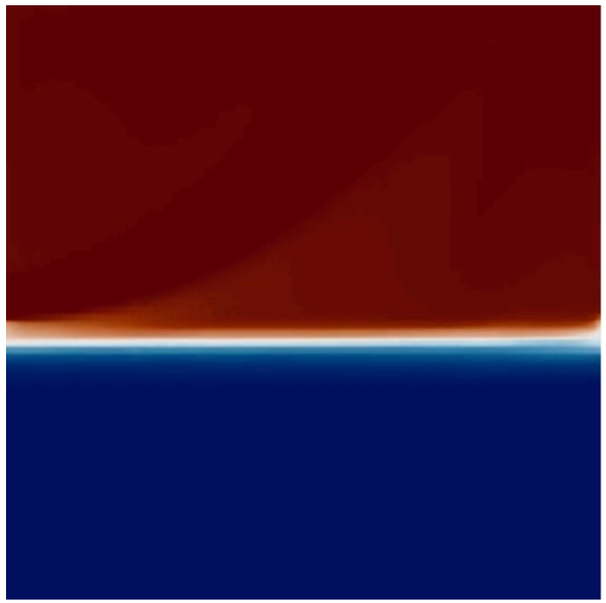
Record the flow once, replay it for the temperature, about *260 times* cheaper.



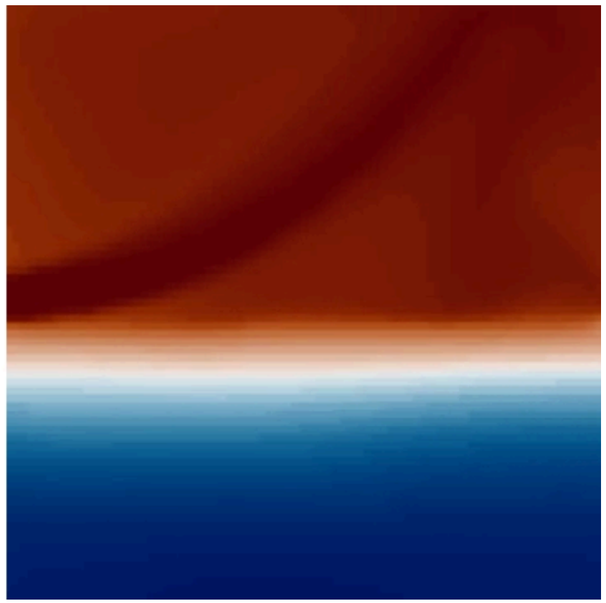
solver time without image output, 569 min against 2.16 min on average; one full CFD run is needed first to record the maps

THESIS FIG 2.1

Same recorded flow, but the replay's front comes out *four times thicker*.



reference CFD



standard replay, $f_{sd} = 0.5$

mid-plane temperature, 293-333 K, decoded from the rendered frames (mean error ≤ 0.5 K)

THESIS CH 5

293 K 313 K 333 K

A warm jet enters a closed tank for 315 s and builds a warm layer over cold water

Laminar and closed: every joule in and out can be accounted for

The front is about 5 mm thick in the reference; the outlet sits just below it

Live playback of decoded rendered frames, one every 2 s

I followed the heat: from the tank budget, to the outlet, to *one face* between two cells.

1 · TANK BUDGET

No mixing strength fits both heat and layer height

best heat error 1.07 K with no mixing; best layer height near f_{sd} = 0.5

2 · OUTLET

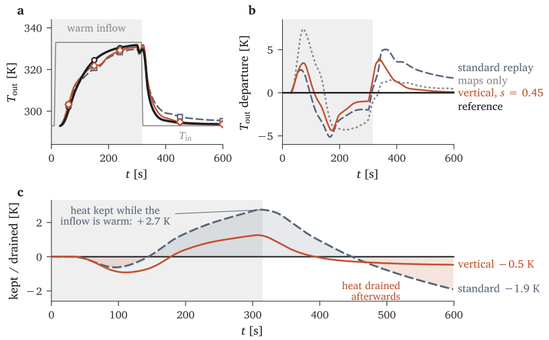
A smeared front drains heat

from the energy balance: 4.6 K of mean temperature leaves after the inflow

3 · ONE FACE

The mixing step, not the recording

recorded maps alone keep the front 2.5 mm; the exchange makes it 19.3 mm



outlet temperature recovered from the energy balance; heat kept while the inflow is warm, drained afterwards

THESIS FIG 5.10

The fix moves heat *within physical bounds*: conservative, and no new extremes.

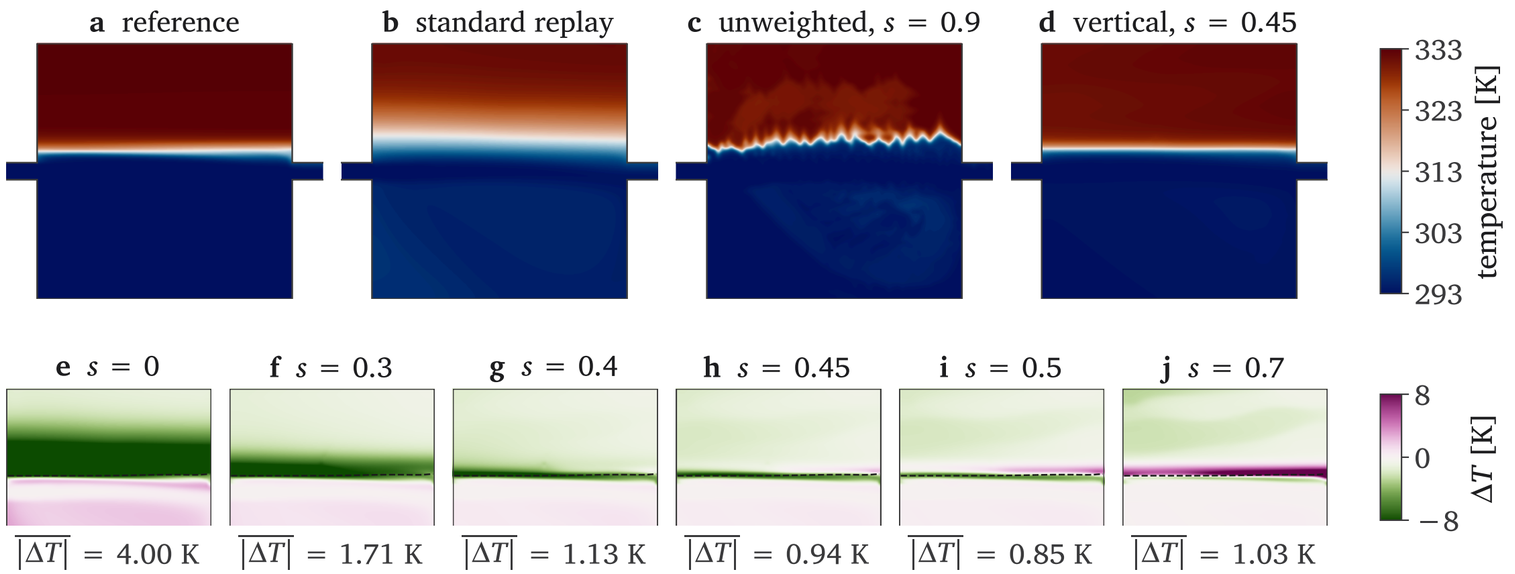
15.7 mm

front, standard replay

≈ 5 mm

front, corrected (5.3–4.5 mm; reference 5.0 mm)

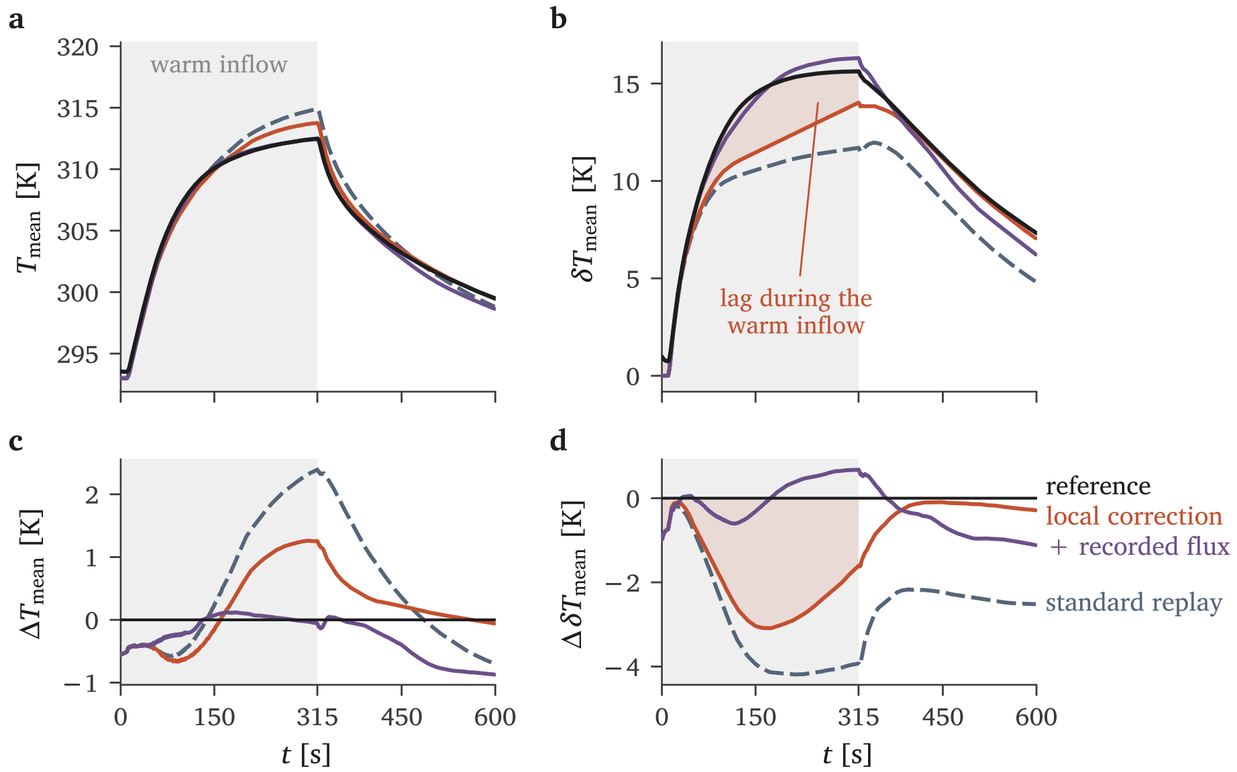
The strength was chosen against the reference. Choosing it without one is the next step.



adiabatic end state and strength ladder, departure from the reference

THESIS FIG 5.9

Knowing *where the method stops* is part of the result.



cooled lid: heat content right, layering late while the tank fills

THESIS FIG 6.3

+0.06 K

end error with the correction, on 9.62 K of cooling

2.25 → 0.45 K

layering lag, without and with a heat flux read from the recording; the tank then ends 0.86 K too cold

What remains is missing transport, and a recording only serves the flow regime it was made in.

Three research directions, each from a *measured weakness*.

UNTESTED

Learn the mixing term

Replace the hand-set exchange strength by a closure learned from recorded data, checked against the reference.

UNTESTED

Regime-aware replay

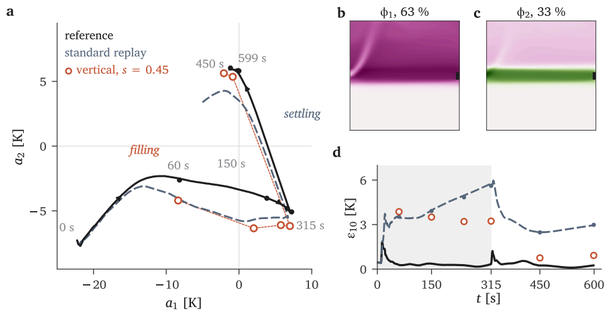
Select or blend recordings by the state of the flow, since one recording serves one regime.

UNTESTED

Physics as acceptance tests

Conservation, boundedness and the energy budget as tests that any fast surrogate must pass.

A hint of low-dimensional structure: in the reference, two POD modes hold 96 % of the mid-plane variance (thesis Fig 5.11).



runs projected on the modes of the reference; the residual tells how far a field lies from the recorded states

THESIS FIG 5.11

Before any industrial use, these points must be shown *on the process's own physics*.

- Reference data per operating regime, and what it costs to produce
- Coverage: does the recording contain the motions the process will see?
- Conservation and boundedness of every correction
- Error on the quantity the process cares about, not only on a test tank
- Comparison with measurements (a lab-rig study is running)
- Physics my tank does not have: turbulence, several phases, reaction heat

1. Follow the energy budget until the error has a location.
2. Fix it within physical bounds: conservative, no new extremes.
3. State where the method stops, with numbers.

Next question: can the replay choose its own correction strength?

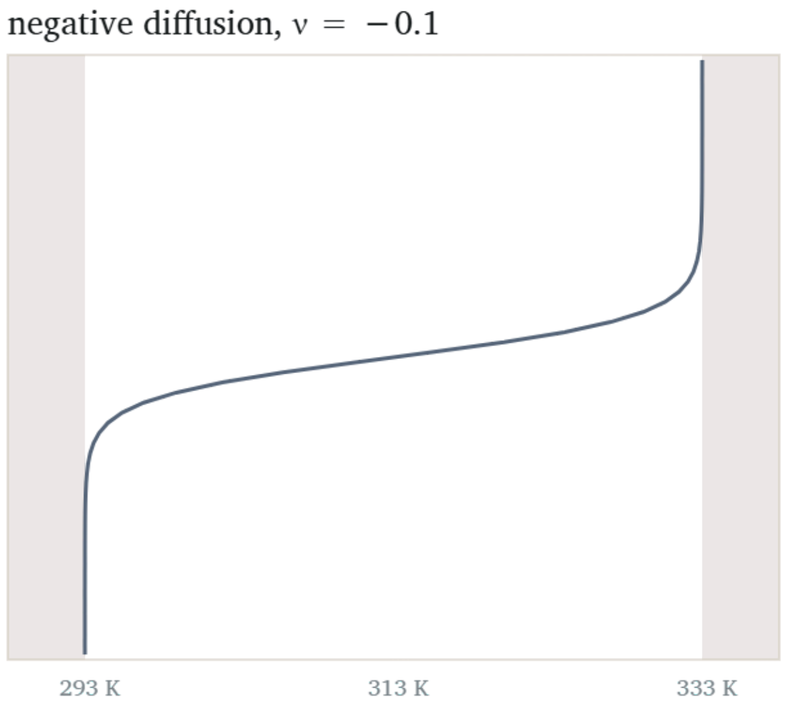
The 260 times is solver time against solver time, without the recording.

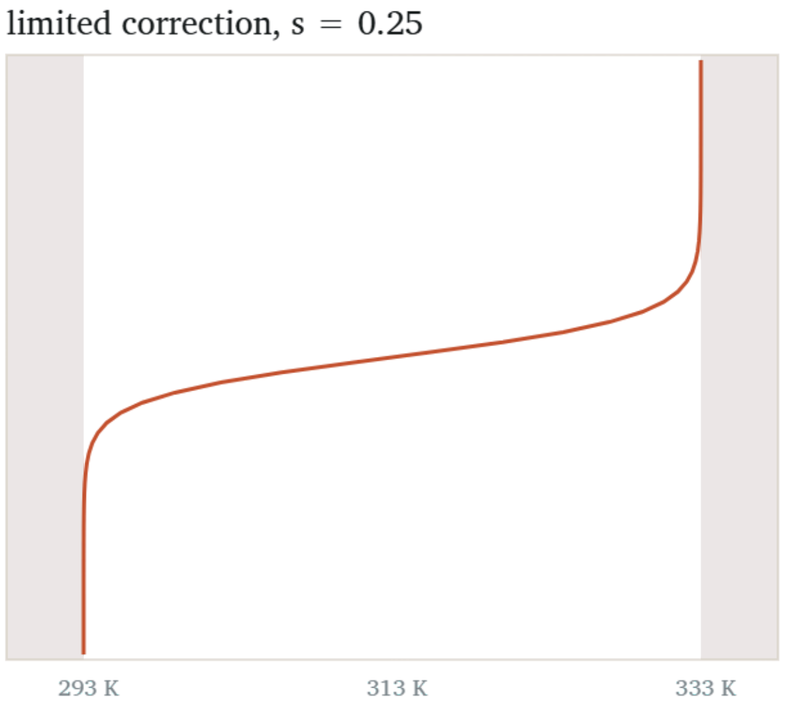
Reference: 569 min for 628 s of flow, image output removed

Replay: 0.81–4.73 min, mean 2.16 min for 600 s; range 120–700x across mixing strengths

The recording is one full CFD run; the saving starts with the second replay

Each face moves at most a share of what both cells can still take.

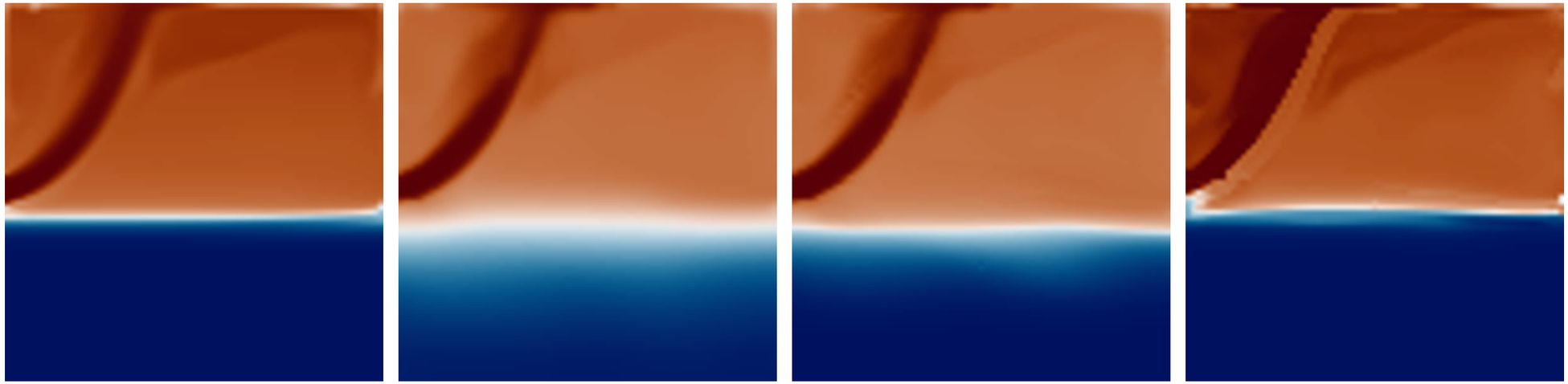




model front: 40 K across 60 cells, ν = −0.1 against limited s = 0.25 (not replay data); shaded: outside the data range

THESIS FIG 4.1

Stored snapshots: reference, standard, corrected, and with the recorded flux.



reference CFD | standard replay | corrected, s = 0.475 | + recorded flux

cooled lid, mid-plane temperature 293-333 K

STORED SOLVER SNAPSHOTS

What I built myself.

- The correction stage in the replay library's user hook: limiter, direction weight, stability criterion
- A recorded convective-flux channel: written during the recording, applied during the replay
- The analysis: energy balances, outlet temperature from the bulk balance, front width, field measures
- Every figure from the solver files, with scripts that recompute the numbers